

6.1 Overview

The research and practice of measurements of the human body is called Anthropometry. Anthropometric measures for the Ghana Breast Health Study include weight, standing and sitting height, hip circumference, and waist circumference. Anthropometric measurements will be made on all consented participants, both Cases and Controls, during the clinic appointment. This chapter discusses how to prepare the participant for the measurements, conduct the measurements, and record the information.

6.2 Equipment and Supplies

The equipment and supplies necessary for anthropometric measurements are as follows:

Equipment

- **Seca 869 Physician Scale:** This digital scale measures to the 0.1 pound or the 0.1 kilogram (kg) and has a capacity of 550 pounds (250 kg).
- **Seca 213 Portable Stadiometer:** This portable stadiometer measures in increments of $\frac{1}{8}$ of an inch or 0.1 centimeters. It has a measurement range of 2 – 90 inches (6 – 230 cm).
- **Seca 201 Ergonomic Circumference Measuring Tape:** This tape measures in increments of $\frac{1}{8}$ of an inch or 0.1 centimeters. It has a measurement range of 15 – 200 cm.
- 15-cm plastic ruler
- Weekly Calibration weights: eleven 10-kilogram weights
- Step stool
- **Adjustable shower stool:** This will be used for the sitting height measurement. For the Ghana Breast Health Study, the grey stoppers at the bottom of each foot of the stool

must be removed and the stool should be set to the tallest height possible when taking the sitting height measurement.

Supplies

- Weekly Calibration Logs for the scale
- AA batteries
- Calculator

6.3 Setup of Equipment

At the beginning of each day, the staff will calibrate the equipment using the following procedures:

Calibration Weights: A supply of eleven 10kg weights is needed for the weekly calibration. Weekly calibration logs are used to record calibration results (Appendix 6-1).

Weight Scale: The Seca 869 Physician Scale should not be moved. Turn the scale on and off by pressing the Start button on the display. Check that the scale is set to kilograms. If it is not, set it to kilograms by pressing ‘kg/lb/st’ button to the right of the Start button.

Stadiometer: The portable stadiometer should be assembled in the study clinic in a place where it will not be disturbed or moved. At the beginning of the day, check that the headpiece moves up and down smoothly. Check that the horizontal bar is firmly attached to the upright sliding section and that the section operates smoothly. If it does not, report this to the Study Manager.

Furnishings: The stadiometer, scale, step stool, and shower stool should remain set up in the clinic. Before the start of each day, check to see that these items are arranged properly and rearrange as necessary.

A second set of identical anthropometry equipment was delivered to PLH in April 2014 for use when completing the study for controls in the field. Setup for field anthropometry is the same as above. The scale, stadiometer, step stool, and shower stool must be placed on a flat, level surface for proper functioning.

6.4 Quality Control

6.4.1 Calibration

The Seca 869 scale is the only piece of equipment that that needs to be calibrated. Scale calibration should be completed on a weekly basis using eleven 10-kilogram weights. The calibration should be conducted as follows:

1. Press the Start button with no load on the scale.
2. Wait for the display to read 0.0.
3. Carefully stack 11 of the 10 kilogram weights on the scale. Make sure the weights stack evenly in the center of the scale. They should not overhang the edges of the scale or lean in any direction.
4. When the scale display has stabilized, record the weight displayed in the column 'Reading (KG)' on the Weekly Scale Calibration Log.
5. Calculate and record the difference between the actual reading (e.g., 110.1 kg) and the expected reading (e.g., 110.0 kg).

It was originally intended that if the difference is < 0.2 kg, the scale could continue to be used and if the difference was 0.2 kg or greater, the scale should be returned to Seca for recalibration. However, since gym weights are being used rather than precise weights manufactured for calibration purposes, a larger difference has been deemed allowable to account for the greater weight variation inherent to gym weights. Upon initial calibration, the actual measurements were found to be consistently: KBTH 108.1 kg, KATH 113.2 kg, and PLH 109.4 kg. These were deemed to be acceptable to NCI.

6. In addition to recording the calibration weight on the Weekly Scale Calibration Log, the Study Manager should report the calibration readings to Westat on a monthly basis. If any change is found from the accepted calibration weights above, the Study Manager should notify Westat immediately.

6.4.2 Equipment Malfunction

6.4.2.1 Scale

1. Digital scales should be handled in a careful manner and should not be dropped, shaken, or thrown around.

2. If the display fails to light up, check the batteries and activate the Start button on the display.
3. If the display fails to read “0.0” and instead “0.1” or “----” appears on the display, switch off the scale with the Start button and start the scale again – there must not be any load on the scale – only its feet should be in contact with the floor.
4. Report any malfunctions of the scale to the Study Manager.

6.4.2.2 Stadiometer

If the headpiece of the stadiometer is not working properly or is damaged, inform the Study Manager.

6.5 General Guidelines

Anthropometric measures will be collected for all study participants during the clinic visit, including pregnant women and women wearing body enhancers (e.g. hip or buttocks) who refuse to take them off for the measurements. The measures should be completed in the order listed below for the smoothest transition between measurements:

- Weight
- Height
 - Standing
 - Sitting
- Circumferences
 - Hip
 - Abdominal (waist)

The guidelines below should be followed when taking anthropometric measurements and subsequently recording the data on the form:

1. **Always tell the participant what you are going to do before you do it.** The Interviewer should explain what she is doing and why, maintaining professionalism at all times.

2. **Turn the participant in the direction needed for a given measure.** This promotes efficiency during the examination by saving time and avoiding unnecessary movement on the part of the examiner.
3. **Position the zero end of the measuring tape below the measurement value.** With respect to circumference measures, the Interviewer should not take any measurement readings with the zero end of the tape placed above the section of the tape with the result.
4. **Avoid parallax when taking measurement readings.** Parallax describes the phenomenon where an observer reads a different value on a measuring device depending on the angle from which it is viewed. The Interviewer should read the measurement with her line of sight directly in front of the value rather than at an angle or from even slightly off to the side. Use a step stool if necessary to read the measure for taller participants.
5. **Record all measurements to the nearest tenth (first decimal place) of a centimeter (0.1 cm).**

6.6 General Guidelines for Reading and Recording Measurements

Taking anthropometric measurements can seem deceptively simple and are often done poorly with errors occurring frequently. At all times, the Interviewer should minimize any measuring errors due to the measuring techniques. Presented below are general guidelines for reading and recording measurements in order to help ensure a high degree of accuracy and reproducible results.

6.6.1 Reading the Measurements

The following are guidelines for reading the stadiometer/tape measure:

1. Numbered metric units are on the stadiometer and tape measure. Each centimeter (cm) is divided into 10 gradations, i.e., small vertical lines that are 0.1 cm (1 millimeter (mm)) apart. The line at 0.5 cm is slightly longer. The tape has both inches and centimeters. For this study, we will use ONLY CENTIMETERS.
2. Be careful to read the stadiometer from top to bottom (or from left to right for the tape measure).
3. If the reading falls between two lines, record the measurements of the LOWER number.

6.6.2 Reading and Recording System

One of the greatest sources of error takes place during the reading and recording of the measurement. To help diminish this type of error, follow these steps after you take a measurement.

1. Immediately call the measurement once out loud. Do not call out the word centimeter. The following examples of how measurements should be called out loud:
 - a. 94.2 should be read as “ninety-four point two”, not “nine four point two”.
 - b. 112.3 should be read as “one hundred twelve point three”, not “one one two point three”.
 - c. Make sure that you read “point” for a decimal point, and make sure that you read zeros. For example, 84.0 should be read as “eighty-four point zero”, not just “eighty-four”.
2. Immediately record the measurement in the boxes provided on the Anthropometry Form (Appendix 6-2). Take your time writing each number clearly.
3. Look at the measurement you have just recorded. Check it to make sure it is accurate and legible.
4. If the woman is pregnant or if the woman is wearing body enhancers (e.g. hip or buttocks) take the measurements and use the comments to record the woman’s pregnancy or enhancers.

It is very important to keep the Anthropometry Form next to you during the measurement process so that you can record the measurement immediately after it is taken. The longer the pause between measurement and recording, the greater the chance of a recording error.

6.6.3 Recording Numbers and Making Corrections

1. Place numbers in appropriate boxes for the measurements. Be careful to make clear and neat numbers the same way every time.
2. Make sure that numbers are written inside the boundaries of the box on the Anthropometry Form.
3. Measurements will be recorded in the form of a 2- or 3-digit number, followed by a decimal point, followed by one digit in the first decimal place.

4. Fill in all boxes. Make sure to put numbers, including zeros, in the correct boxes.
5. If a measurement cannot be obtained for any reason, describe why the measurement wasn't taken in the last box to the right of the measurement.
6. If you make a mistake cross out the mistake using one horizontal line through the ENTIRE number. Then, rewrite the measurement below the boxes.

6.6.4 Replicate Measures

Perform each anthropometric measurement at least twice on each study participant to minimize error. When taking multiple measurements, it is important that each set of measurements be as independent as possible. This means that the Interviewer should not look at the first value when reading and recording the second measurement. Doing so might influence the Interviewer's second measurement, and would defeat the purpose of taking two measurements. The Interviewer should explain to the participant that she will take all the measurements once and then again to ensure accuracy.

6.6.5 Weight

Participants will be weighed in kilograms using the Seca 869 Physician Scale.

6.6.5.1 Preparation

Ask the participant to remove her shoes (but not socks). If the participant does not have any socks on, place a piece of clean tissue paper or a paper towel on the floor next to the scale, if possible, for the participant to step on during weight measurements.

6.6.5.2 Measurement Procedures

Press the Start button to turn the scale on and wait until the display reads "0.0 kg." After introducing the exam to the participant, direct her to stand in the center of the scale platform, place her hands at the sides, and look straight ahead (Exhibit 6-1).

After the participant is correctly positioned and the readout on the scale becomes stable, read the weight and record it on the Anthropometry Form. Ask the participant to step off the scale. The scale will hold the weight value until it is tapped to release the value.

Repeat the weight measurement a second time according to the procedure described above and record it on the Anthropometry Form.

Exhibit 6.1. Weight



6.6.6 Standing Height

Standing height is an assessment of maximum vertical size. Standing height is measured with a stadiometer with a moveable headboard.

6.6.6.1 Preparation

The participant should already have removed her shoes for the weight measurement. If the participant has hair ornaments, jewelry, buns, braids, or corn rows on the top of her head that may interfere with the accurate measure of height, ask the participant to remove them, but do not insist. If the participant refuses to remove her shoes or hair ornaments, please follow the height adjustment procedures in Section 6.6.6.3 below.

6.6.6.2 Measurement Procedures

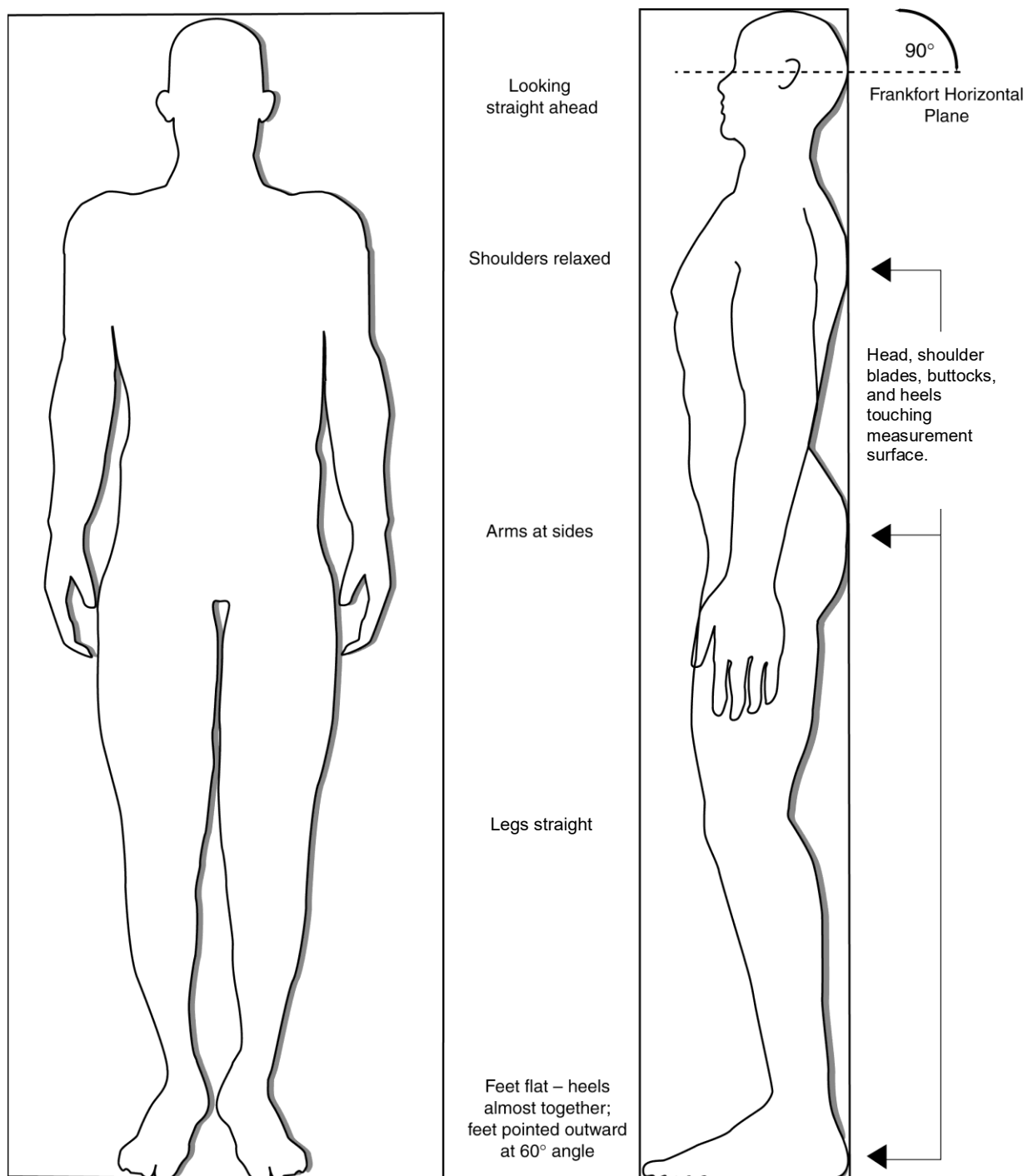
Have the participant stand on the floor in front of and directly below the stadiometer with the heels of both feet together and the toes pointed slightly outward at approximately a 60° angle (see Exhibit 6-2). Make sure the body weight is evenly distributed and both feet are flat on the floor. Check the position of the heels, buttocks, shoulder blades, and the back of the head for contact with the wall. Depending on the overall body conformation of the individual, all points may not touch. Ask the participant to move back against the wall until either the back or buttocks touch, whichever touches first. Make sure the participant's trunk is vertical above the waist and that the arms and shoulders are relaxed.

Align the head in the Frankfort horizontal plane. The head is in the Frankfort plane when the horizontal line from the ear canal to the lower border of the orbit of the eye is parallel to the floor and perpendicular to the vertical backboard. Many people will assume this position naturally, but for some it may be necessary to make a minor adjustment. If required, gently tilt the head up or down until proper alignment is achieved with eyes looking straight ahead. Once correctly positioned instruct the participant to take a deep breath and stand as tall as possible. A deep breath will allow the spine to straighten, yielding a more consistent and reproducible stature measurement. Position the headpiece firmly on top of the head with sufficient pressure to compress the hair. Read the measurement to the nearest 0.1 centimeter, and enter the measurement on the Anthropometry Form.

6.6.6.3 Height Adjustments

If the participant has hair ornaments that cannot be removed that interfere with the accurate measure of height, an adjustment needs to be made to correctly enter height on the Ghana Breast Study Anthropometry Form. Using the 15-cm plastic ruler measure the height of the hair ornament and subtract this number from the total number read on the Stadiometer. Then enter this new number on the Ghana Breast Study Anthropometry Form. This adjustment will be entered for both the standing and sitting height.

If the participant refuses to remove her shoes, measure the height of the shoe heel using the 15-cm plastic ruler and subtract this number from the total number read on the Stadiometer. Then enter this new number on the Anthropometry Form.

Exhibit 6-2. Participant position for standing height

6.6.7 Sitting Height

Sitting height is an assessment of the length of the trunk. Sitting height is measured with a stadiometer that has a moveable headboard and a shower stool without back support.

6.6.7.1 Preparation

Similar to standing height measurements, the participant should already have removed her shoes. If the participant has hair ornaments, jewelry, buns, braids, or corn rows on the top of her head that may interfere with the accurate measure of height, ask the participant to remove them, but do not insist. If the participant refuses to remove any ornaments from the back or top of her head, then the height adjustment ruler can be used.

Move the sitting height shower stool directly in front of the stadiometer. The grey rubber feet at bottom of each foot of the shower stool must be removed in order for it to fit over the base of the stadiometer. The shower stool should be set to the tallest height possible when taking the sitting height measurement.

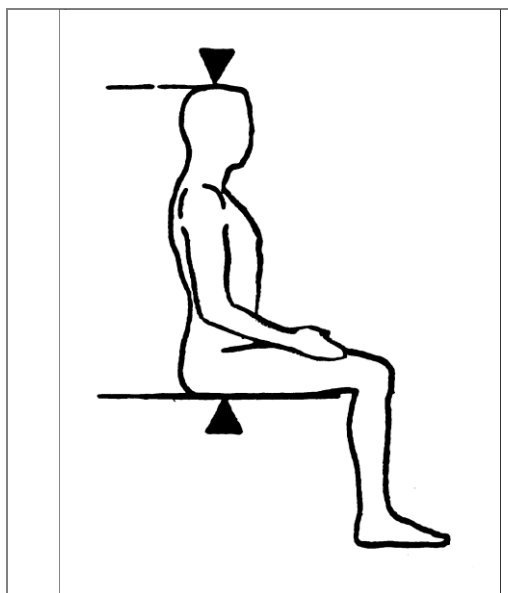
6.6.7.2 Measurement Procedures

Have the participant sit on the seat with her back against the stadiometer. The participant should be positioned so that her back and buttocks are flush with the measuring rod of the stadiometer, while sitting as upright as possible with her head in the Frankfort Horizontal Plane. Her knees should be directed straight ahead with arms and hands resting at the sides. Make sure the body weight is evenly distributed and both feet are flat on the floor. Check the position of the buttocks, shoulder blades, and the back of the head for contact with the wall. Depending on the overall body conformation of the individual, all points may not touch. Ask the participant to move back against the wall until either the back or buttocks touch, whichever touches first. Make sure the participant's trunk is vertical above the waist, and that the arms and shoulders are relaxed.

Align the head in the Frankfort horizontal plane. If required, gently tilt the head up or down until proper alignment is achieved with eyes looking straight ahead. Once correctly positioned instruct the

participant to take a deep breath and sit as tall as possible (see Exhibit 6-3). Position the headpiece firmly on top of the head with sufficient pressure to compress the hair. When the participant is properly positioned, read the measurement to the nearest 0.1 centimeter, and enter the measurement on the Anthropometry Form.

Exhibit 6-3. Participant position for sitting height



6.6.8 Hip Circumference

Hip circumference is a measurement made horizontally at the level of maximum extension of the buttocks posteriorly. Hip circumference is measured with the Seca 201 measuring tape.

6.6.8.1 Preparation

For participant modesty and comfort, participants are not asked to lower their clothing to measure this circumference. Measurements are taken over the participant's clothing.

6.6.8.2 Measurement Procedures

Squat on the participant's right side and extend the tape measure around the participant's buttocks (over her clothes). Position the tape in a horizontal plane at the maximum extension of the buttocks (see Exhibits 6-4 and 6-5). If available, use a wall mirror to ensure correct alignment of the tape. Ensure that the tape lies parallel to the floor and remains snug but does not compress the skin. Read the measurement from the right side of the body at the mid-axillary line and record on the Anthropometry Form.

Exhibit 6-4. Diagram of site for hip circumference

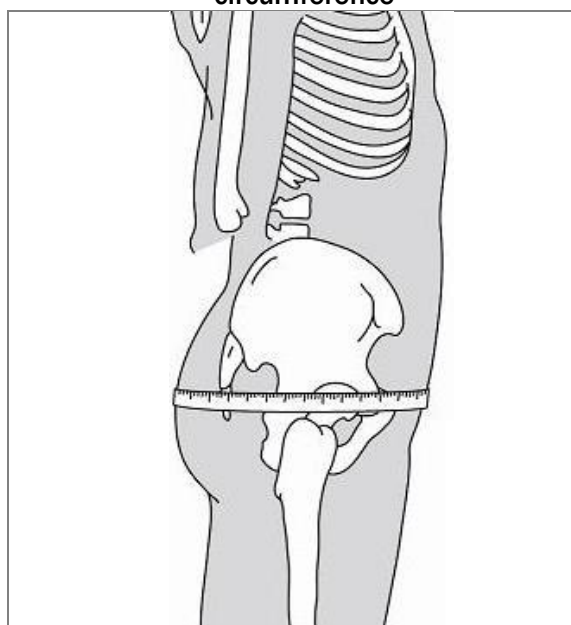


Exhibit 6-5. Measurement site for hip circumference



6.6.9 Waist Circumference

Waist circumference is a measure of the abdomen at a specific anatomical landmark. Waist circumference is measured with the Seca 201 measuring tape.

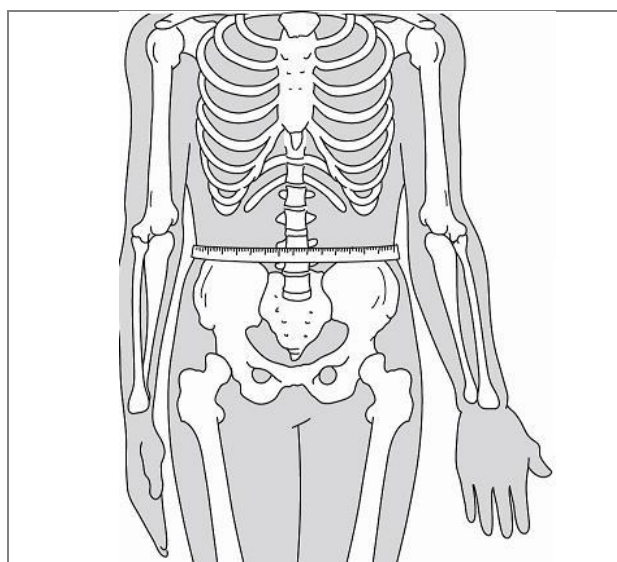
6.6.9.1 Preparation

For this study, women are likely to be wearing clothes that will not make it possible to take the measurement against the participant's skin. For consistency across all women, the waist circumference measurement will be taken over the participant's clothing.

6.6.9.2 Measurement Procedures

Instruct the participant to point to her navel to identify where the waist circumference measurement should be taken. Squat on the participant's right side and extend the tape measure around the participant's waist (Exhibit 6-6), over the participant's navel. If available, use a wall mirror to ensure correct alignment of the tape. Ensure that the tape lies parallel to the floor and remains snug but does not compress the skin. Read the measurement from the right side of the body and record on the Anthropometry Form.

Exhibit 6-6. Diagram for measurement of waist circumference



6.7 Performing Replicate Measurements

Perform each anthropometric measurement again in the same order they were done the first time to minimize error. The Interviewer should not look at the first set of measurements when recording

the second measurement. The Interviewer should explain to the participant that she will take all the measurements again to ensure accuracy.

After the second set of measurements are taken, compare the two measurements and determine if they are inside the acceptable range for replicate measurements. If the difference between the first two readings is greater than 0.3kgs for weight or 1cm for all other measurements, then take a 3rd reading. Take the average of the two closest readings and record this value on the Anthropometry Form. If a measurement is not taken, for reasons such as equipment failure, participant refusal, or any other reason, then record this on the Anthropometry Form.

6.8 Care and Maintenance

The anthropometry equipment requires minimal maintenance; however, all equipment should be monitored regularly for proper maintenance and functioning. All equipment should be cleaned regularly using a germicide. In addition, the surface of each piece of equipment that comes into contact with the participant should be cleaned prior to and following each visit.